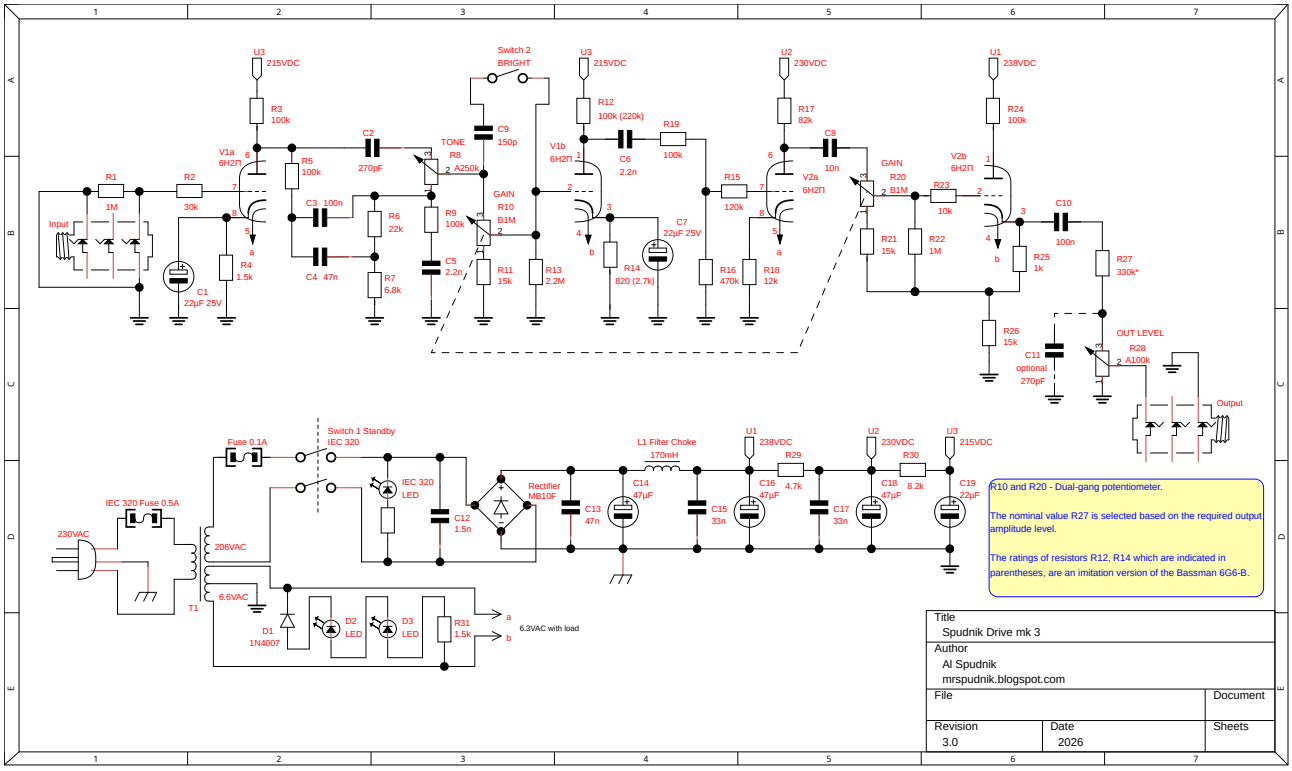
Spudnik Drive Mark III
by Al Spudnik



The design is positioned as a preamplifier for studio recordings and connection to a speaker simulator or power amplifier.

You can connect a guitar directly to the input of this preamplifier, as well as a processed signal - for example, a tube reverb with a classic 6G15 circuit, or its FET copies. To add variety to overdriven amp, you can connect absolutely any booster, treble booster, overdrive pedals like the Tube Screamer, DOD 250, or even a fuzz box to the input. By manipulating the TONE, BRIGHT, GAIN knobs, you can achieve an endless variety of overdrive.

Design and components:

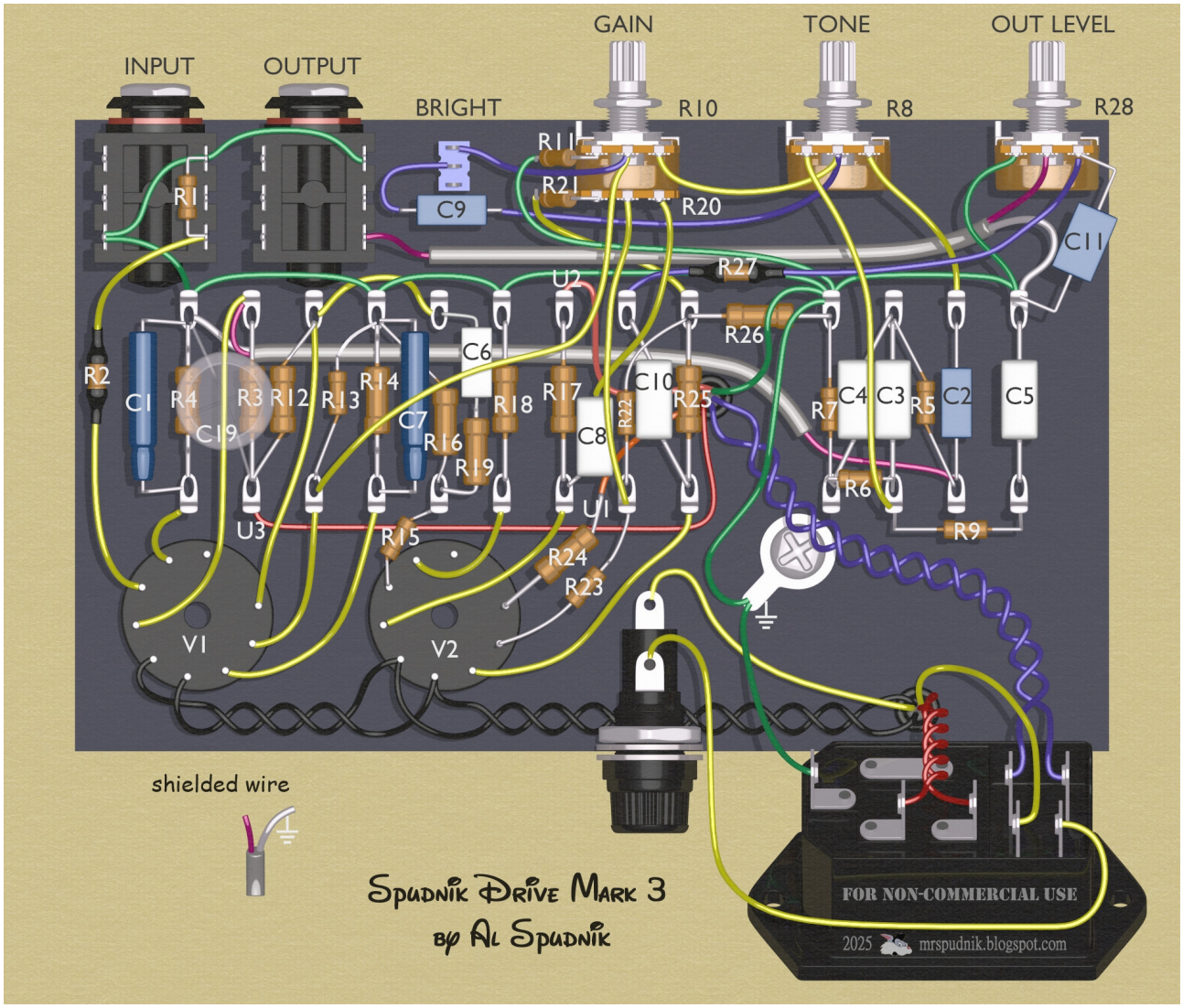


I did use soviet 6N2P (6Н2П). You can use any analogs, such as ECC83, 7025, 5751, or 12AX7 (take note pinout!). In this circuit, 6N2P tubes operate at a lower anode voltage. This, too, is up to you to decide how close to the original Showman or Bassman voltage you want it to be.

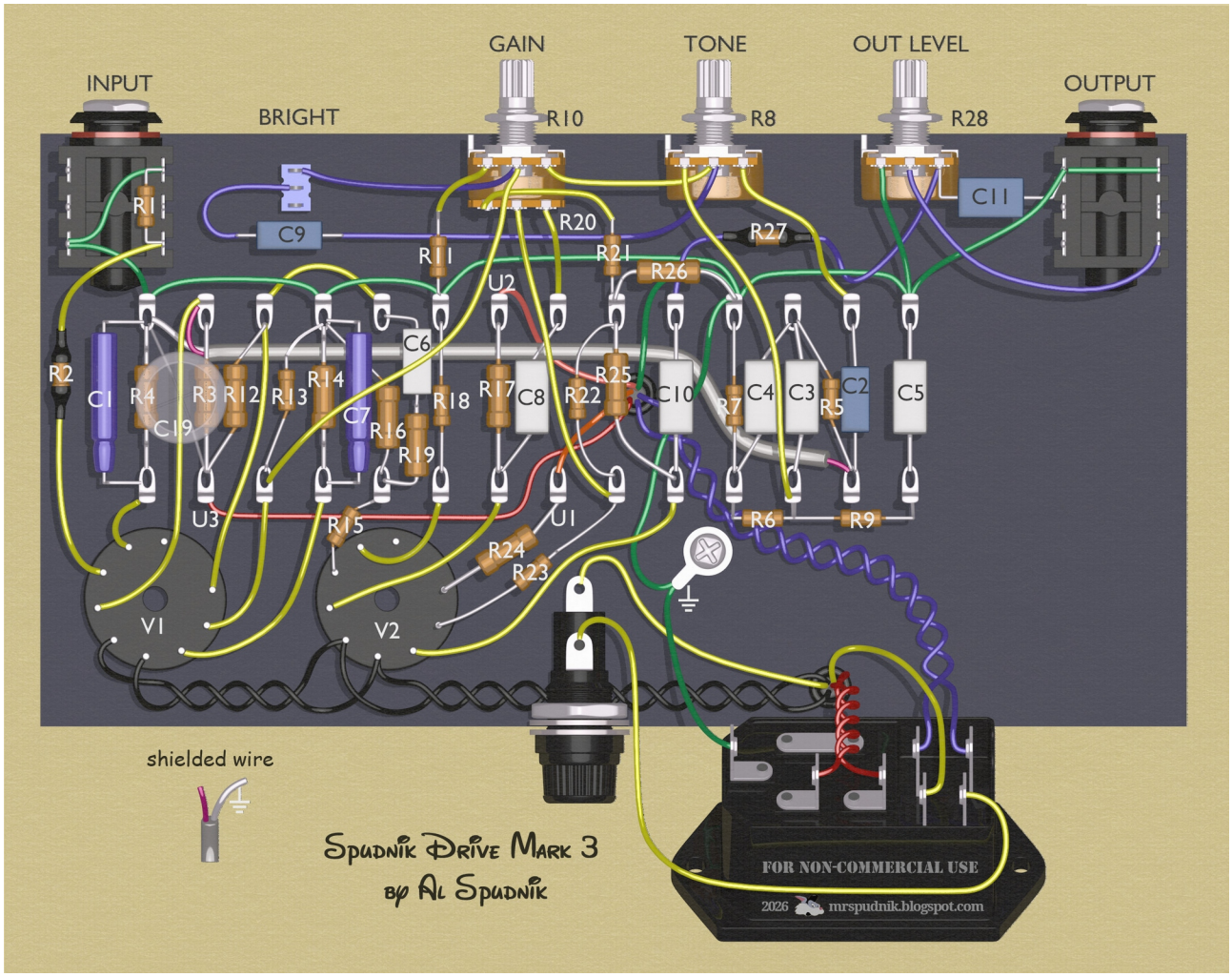
Use high-quality audio-grade film capacitors, such as Evox Rifa Metal Film, with a minimum voltage rating of 400 volts for C2, C3, C4, C5, C6, C8, C9, and C10 (with the anode voltage increased to Fender levels, the capacitors are rated for 630 volts). C1 and C7 are high-quality audio-grade electrolytic capacitors, such as Nichicon, with a minimum voltage rating of 25 volts.

As can be seen in the diagram, the lamp preheating is turned on immediately when the device is connected to the network, and the switch of the IEC320 AC-17 socket is connected as «Standby».

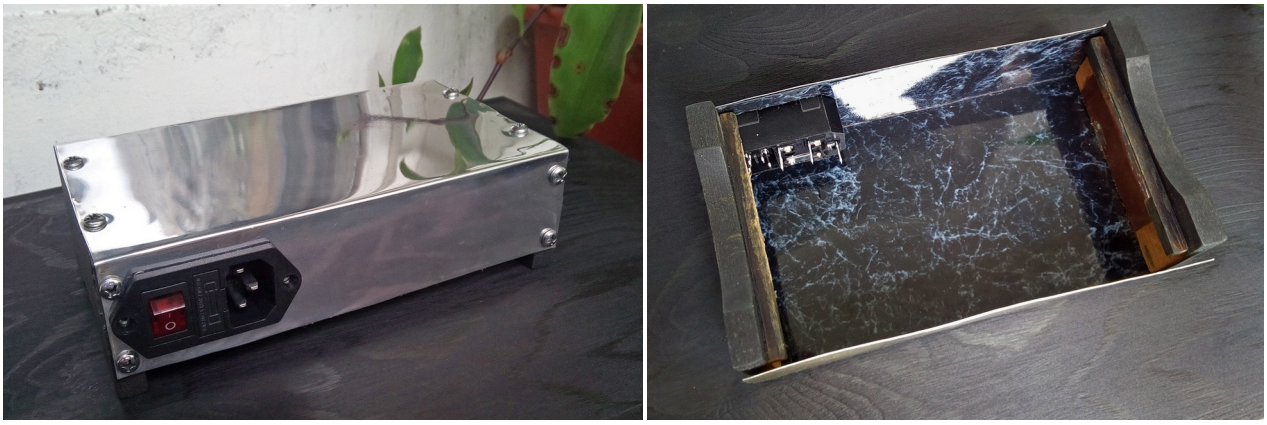
The power transformer must have a shielding layer between the mains winding and the anode winding. Or the filament winding - with a center tap connected to ground - should be wound between them.



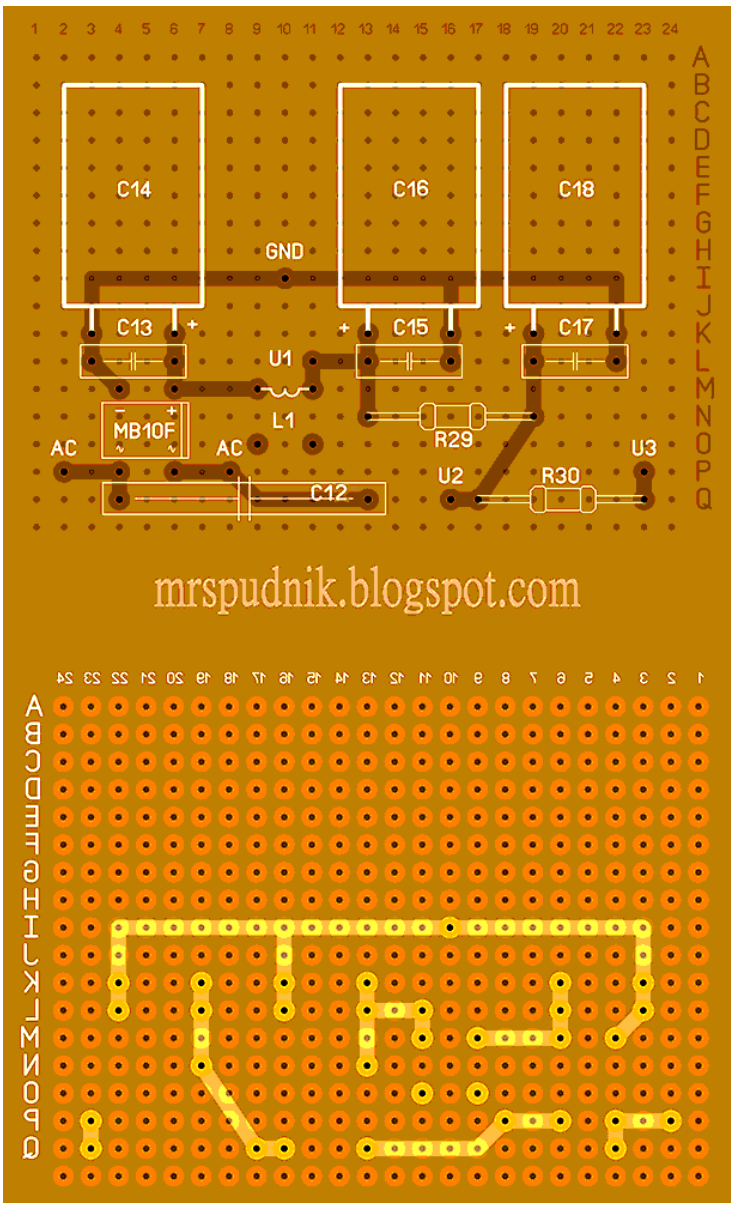
I offer two versions of the layout. The one I did, with the jack sockets next to each other.



And a more advanced version - with input and output on the edges of the front panel.



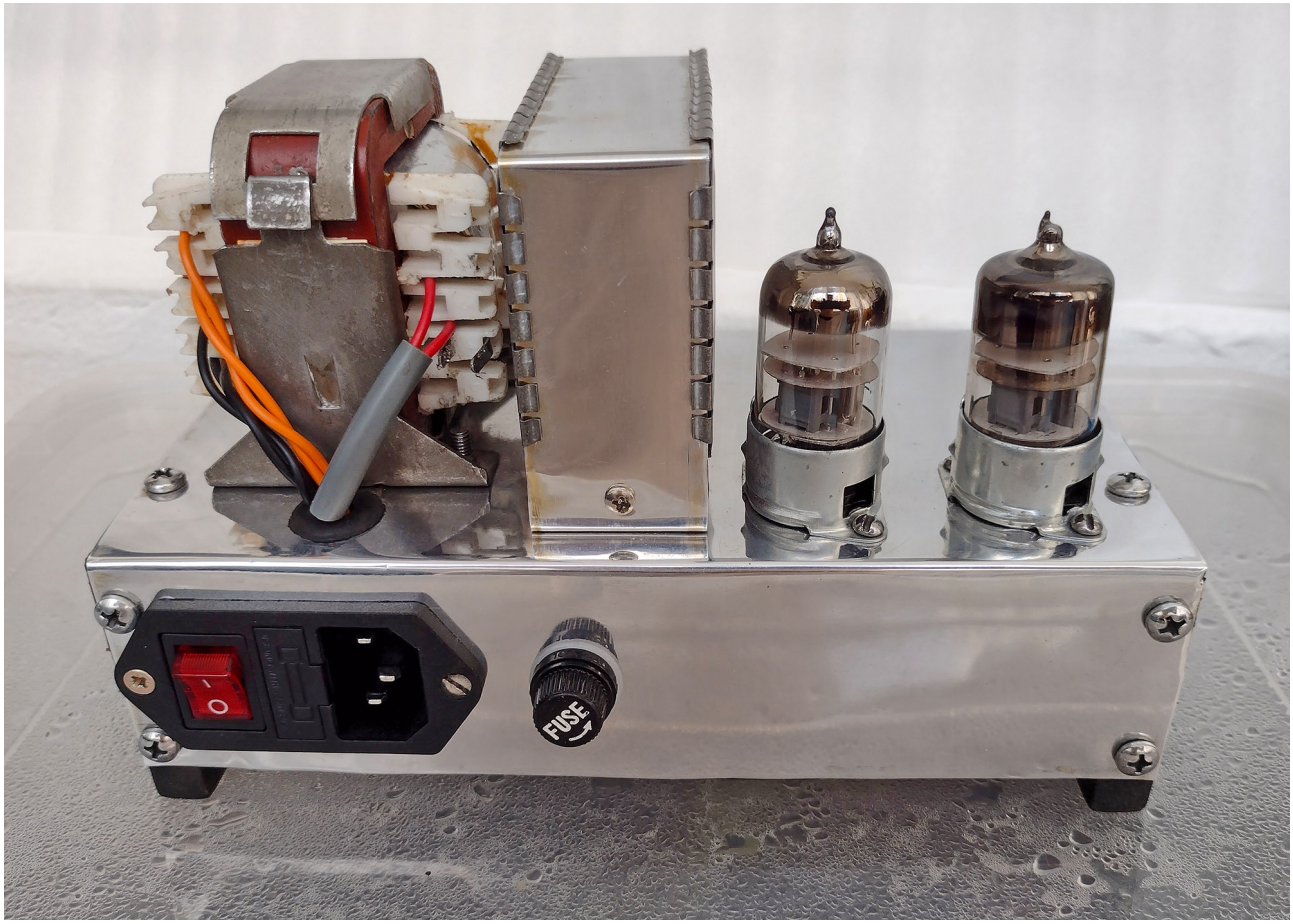
The chassis is made of bent sheet aluminum (0.33 mm thick) bonded to a thin plastic sheet and secured at the sides to wooden supports.



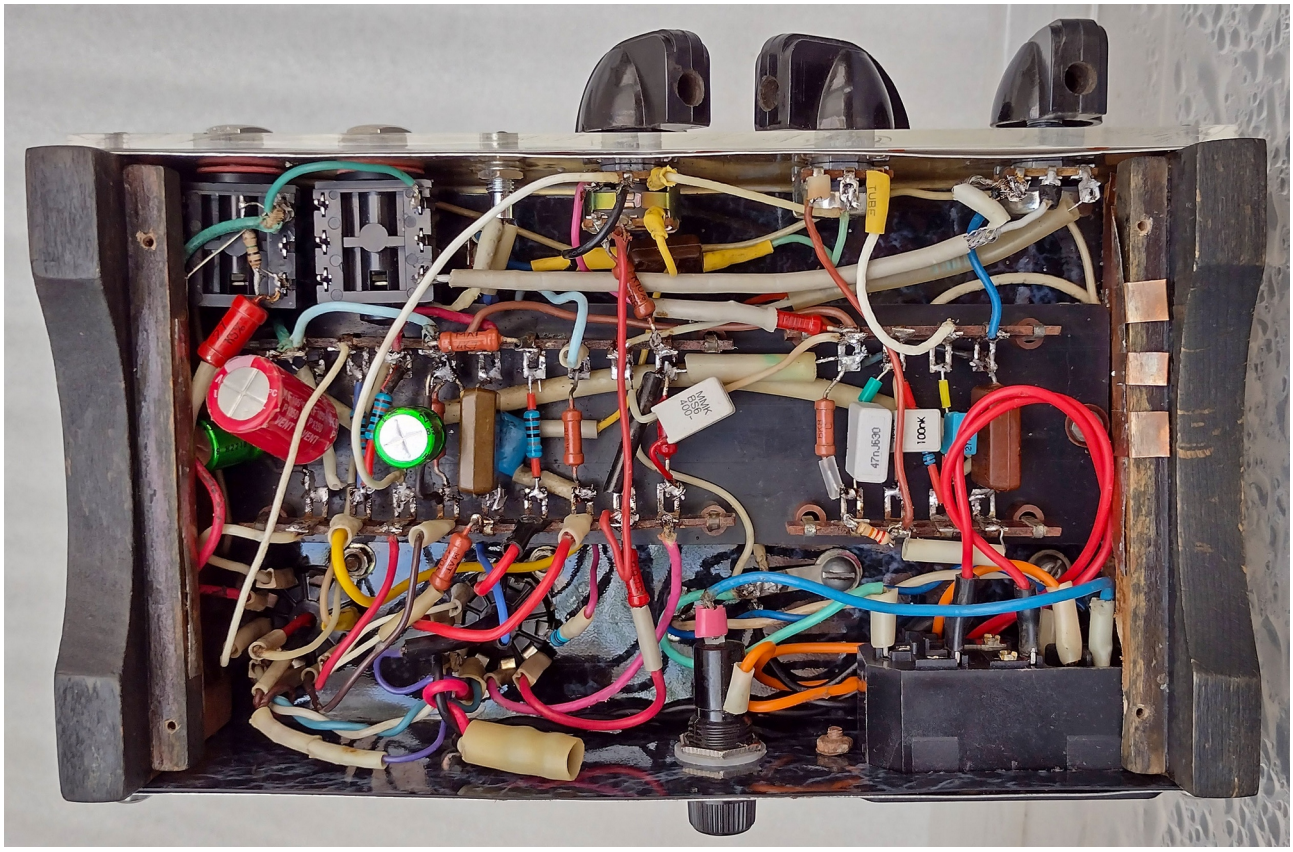
Voltage rectifier, a choke with a filter circuit made on a perforated board measuring 50/70 mm.

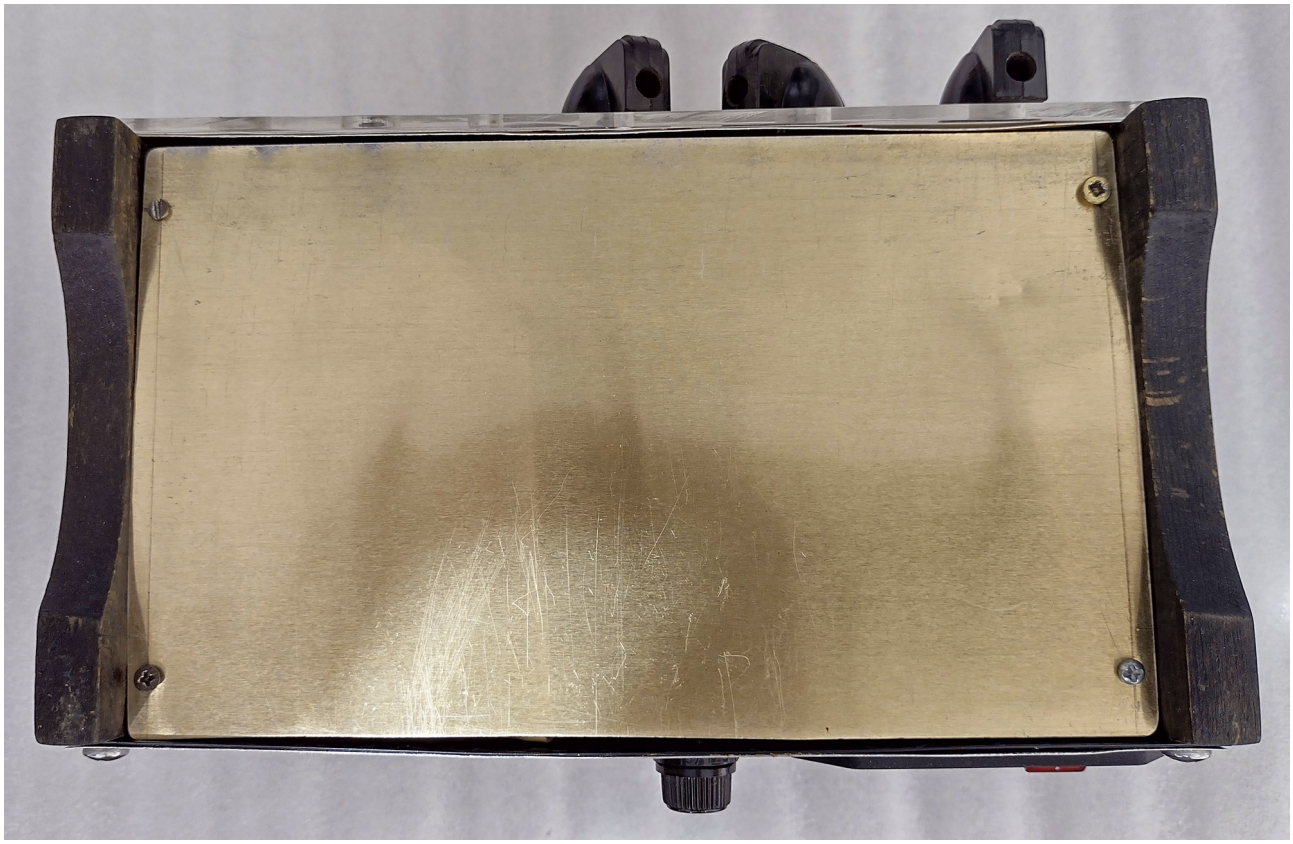


The board is located on the chassis in a shielded housing.


FUSE


Spudnik Drive
INPUT
OUTPUT


MARK
1986
2000
47n/630
100K



All other components are located inside the chassis on mounting rails. The aluminum bottom cover provides complete shielding.

Parts List:

Resistors (metal film)

R1, R22 = 1M (0,25W)
R2 = 30k (0,25W)
R3, R12, R19, R24 = 100k (0,5W)
R4 = 1.5k (0,5W)
R5, R9 = 100k (0,25W)
R6 = 22k (0,25W)
R7 = 6.8k (0,25W)
R11, R21 = 15k (0,25W)
R13 = 2.2M (0,25W)
R14 = 820 Ohm (0,5W)
R15 = 120k (0,25W)
R16 = 470k (0,5W)
R17 = 82k (0,5W)
R18 = 12k (0,5W)
R23 = 10k (0,25W)
R25 = 1k (0,5W)
R26 = 15k (0,5W)
R27 = 56-330k (0,25W)
R29 = 4.7k (1W)
R30 = 8.2k (1W)
R31 = 1.5k (0,25W)

Potentiometers

R8 = A250k (logarithmic)
R10, R20 = B1M (dual-gang linear)
R28 = A100k (logarithmic)

Capacitors

C1, C7 = 22µF 25V (Audio Grade Electrolitic)
C2, C11 = 270pF 400V (Evox Rifa PFR5 MKP, or similar Metal Film)
C3, C10 = 100nF 400V (Evox Rifa MMK, or similar Metal Film)
C4 = 47nF 400V (Evox Rifa MMK, or similar Metal Film)
C5, C6 = 2.2nF 400V (Evox Rifa PFR, or similar Metal Film)
C8 = 10nF 400V (Evox Rifa MMK, or similar Metal Film)
C9 = 150pF 400V (Metal Film or Ceramic)
C12 = 1.5nF 630VAC (Polypropylene Film)
C13 = 47nF 400V (General Purpose Film Metallized Polypropylene)
C14, C16, C18 = 47µF 400V (General Purpose Electrolytic)
C15, C17 = 33nF 400V (General Purpose Film Metallized Polypropylene)
C19 = 22µF 400V (General Purpose Electrolytic)

Diodes

D1 - 1N4007 or another

D2, D3 - 3mm LED
Rectifier MB10F

Tubes
V1, V2 - 6N2P
Two nine-pin tube sockets

Sockets
Two jack sockets 6.35mm female with switch
Power socket IEC320 AC-17

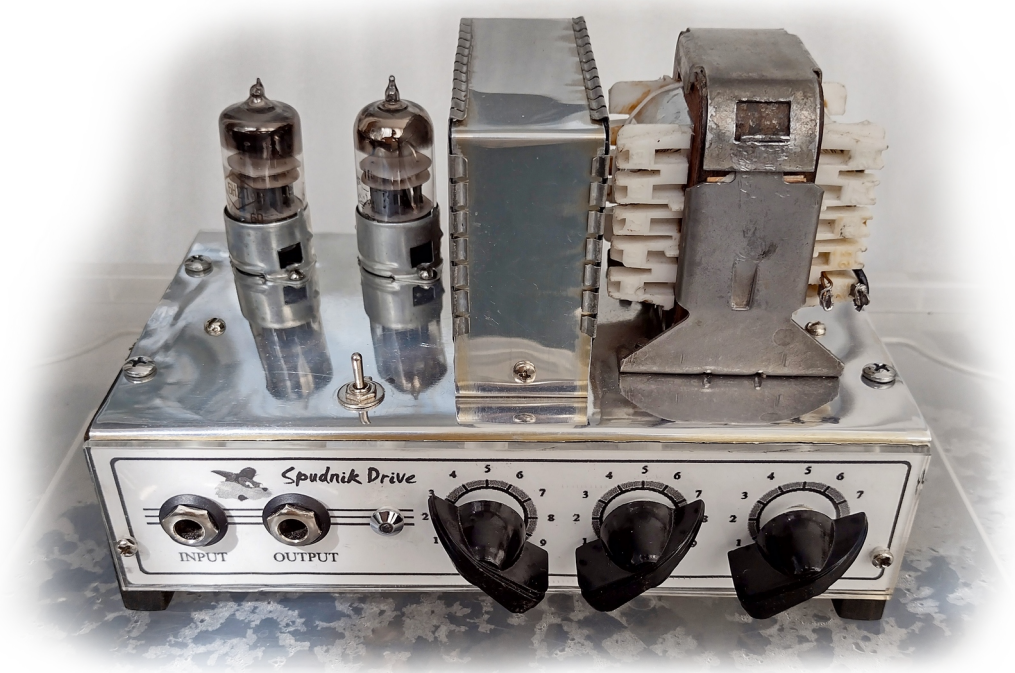
Switches

Switch 1 - IEC320 AC-17 Rocker Switch Power Socket with LED

Switch 2 - Mini Toggle Switch SPDT on-off

Power Transformer
Power 15...20W. Anode winding = 200...210v AC, 10mA minimal. Filament winding = 6.6v AC, 0.8A.

Filter Choke
Inductance = 0.17 Henry, Current = 10mA



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